AIRFARE MARKETS UNDER PRESSURE: WHEN COMPETITION FAILS, TRAVELERS PAY

SDSS DATATHON 2026

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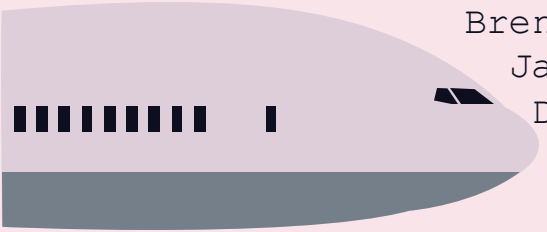

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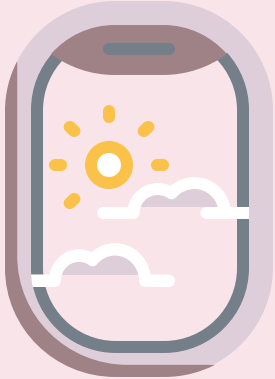

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01 PROBLEM OVERVIEW

The Conflict for customers
Finding the best deal when flying is hard. A trade-off between price and flight comfort forces many consumers to make hard choices, often without even considering the best one.

The Gap
Families and students are often priced out of "Fortress Hubs" where a single carrier dictates rates regardless of flight miles.

The Objective
Use machine learning to move beyond anecdotal travel advice and quantify the structural cost of market concentration.

The Goal
Identify where competition is failing to inform better booking for travelers and smarter oversight for policymakers.

02 OUR APPROACH ON DATA

EDA

Firstly, we used preliminary Exploratory Data Analysis (**Pandas & NumPy** for data manipulation and **Matplotlib & Seaborn** for trend visualization and regression analysis), to discover basic trends on the data.

MODELS

We created 3 models to simulate the current pricing system: **#1: Baseline model** with no tuning. **#2: Random Forests model** with tuning. #3: **Gradient Booster Model** with tuning.

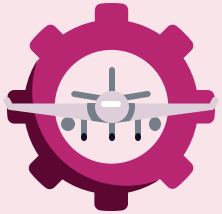
INSIGHTS

Model # 1 and # 2 provided insights for consumers and airlines, while model #3 gave valuable info for policymakers regarding market share.

EFFECT

Combining these results with **SHAP**, scatter plots, and simple regression, we determined each individual variable's effect on airfare.

CONSIDERATIONS ON DATA



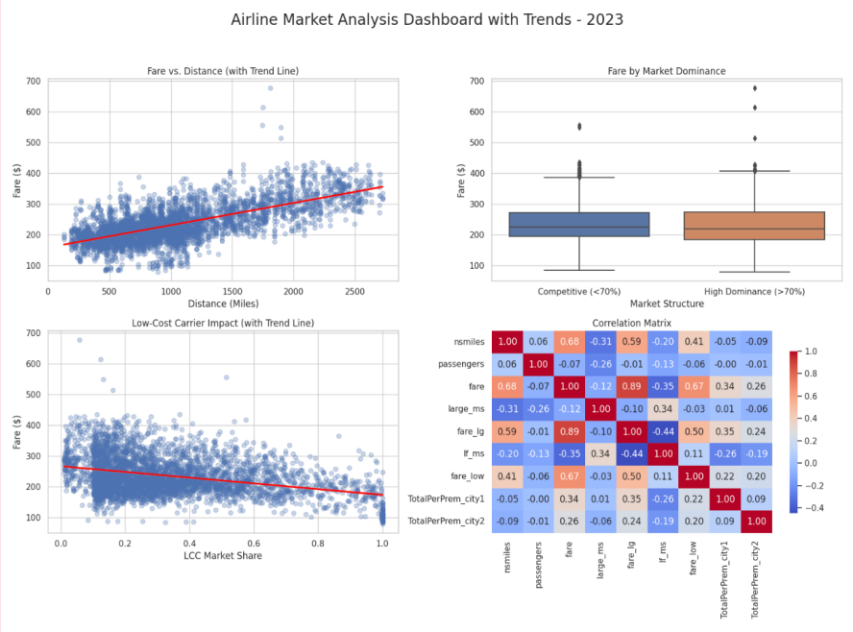
- City Market ID (Endpoint 1 and 2) where dropped from the analysis since we decided to use city name to represent travel locations.

- We divided the analysis into 2: First considering relevant variable's effect on airfare, and then the direct effect of market power into price.
    - This would allow us to provide separate insights to customers and airlines (first part=, and for governments (second part, by helping regulate)

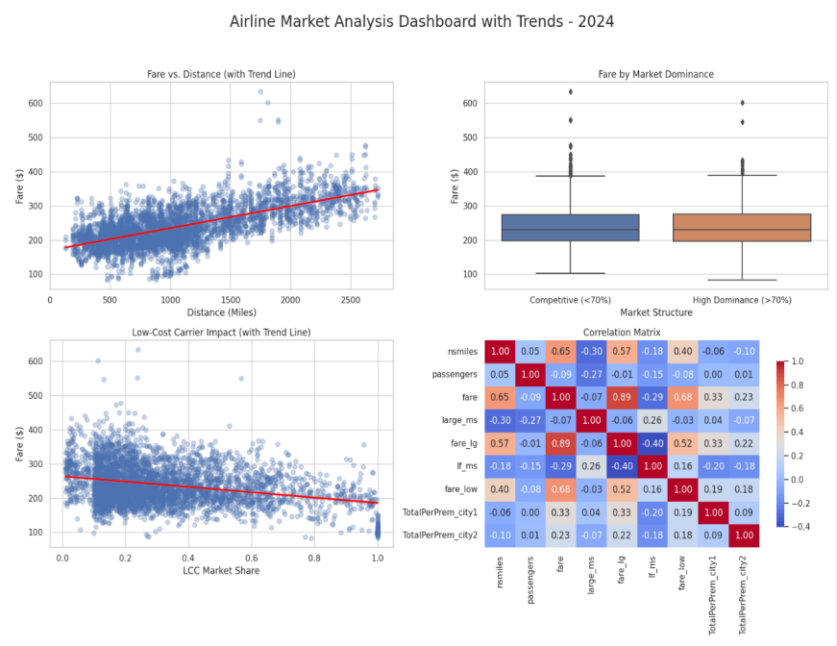
03 TRENDS AND PREDICTIVE MODEL

Performing regression analysis, we found the following correlation between variables and airfares for the following years. 2022 was excluded since Covid effects greatly affected data.

All quarters 2023

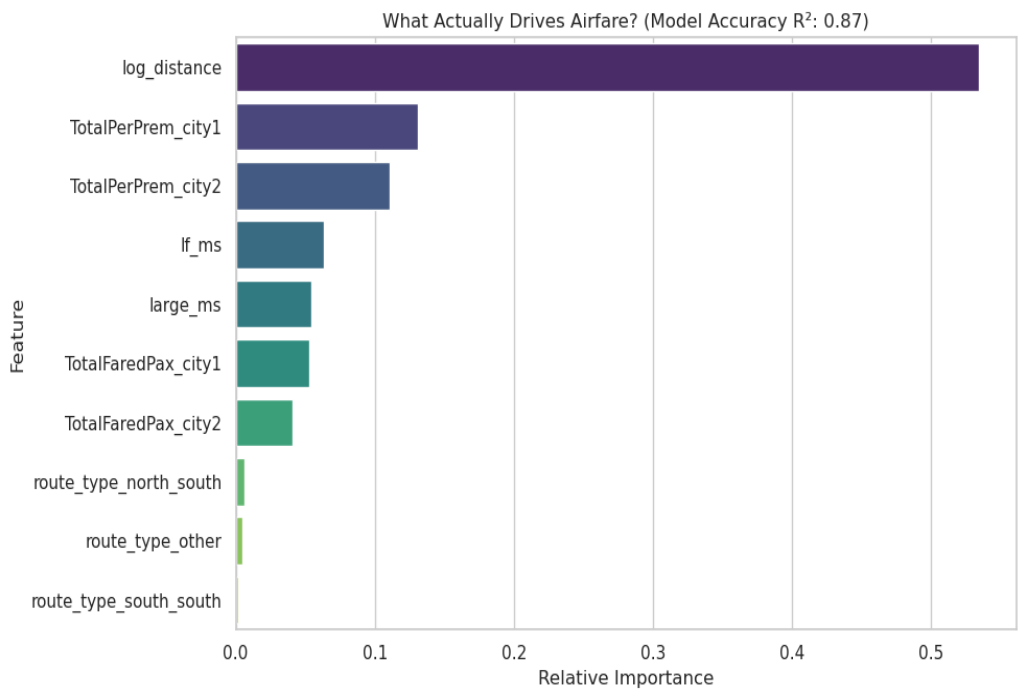


All quarters 2024



OUR PREDICTION MODEL

Model # 2: Random Forests with tuning



- We used log(distance), based on our findings that after long distances, the marginal effect becomes low.
- All flights were classified into 3 aoute types:
  - Route_type_north_south: Flights from a northern city to a southern, or visceversa.
  - Route_type_south_south: Flights from southern to southern city.
  - Route_type_other.
- MAE of $17.48, and adjusted R^2 of 0.8691.

04 THREE KEY FACTORS FOR AIRFARES

FLIGHT DISTANCE

(Route Characteristic)

- Baseline model $R^2 = 0.4822$. Distance explains ~48% of fare variation, but this correlation is weakening year-over-year (from 0.654 in 2022 to 0.501 in 2025). Physical geography is losing predictive power.

CITY COMPETITION

(Hub Dominance Effect)

- Cities with dominant carriers (>70% market share) show significant fare premiums; Charlotte (+15.8%) and Atlanta (+12.4%) command higher prices.
- Whereas cities with LCC presence show systematic discounts. Tampa (-32.4%) and Las Vegas (-20%) offer the deepest savings. Where competition thrives, prices fall for everyone.

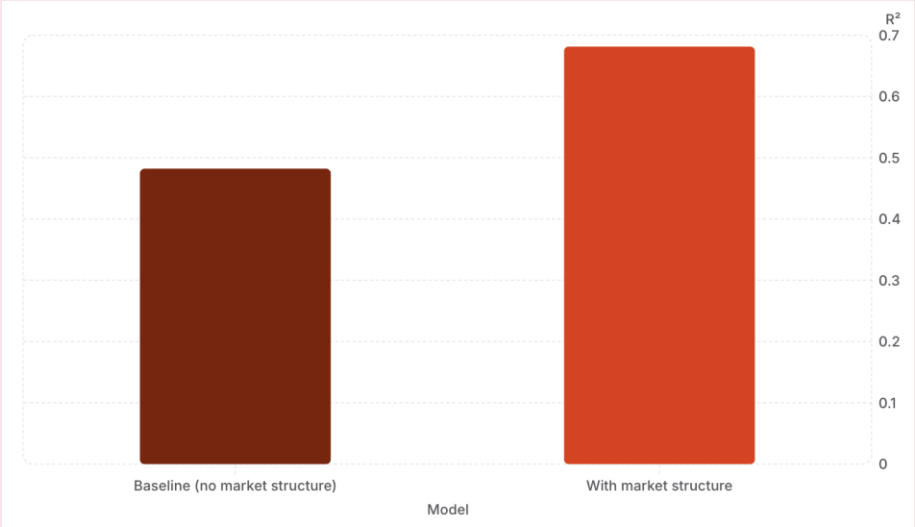
TIME OF THE YEAR

(Seasonal Shifts)

- Airfares typically reach their lowest point in quarter 3 each year, as the post-summer travel slump leads to decreased demand and lower prices.
- Conversely, Q2 often sees peak pricing driven by spring break and the start of summer vacation surges.

MARKET STRUCTURE EFFECTS. DOES IT ACTUALLY AFFECT THE FARE?

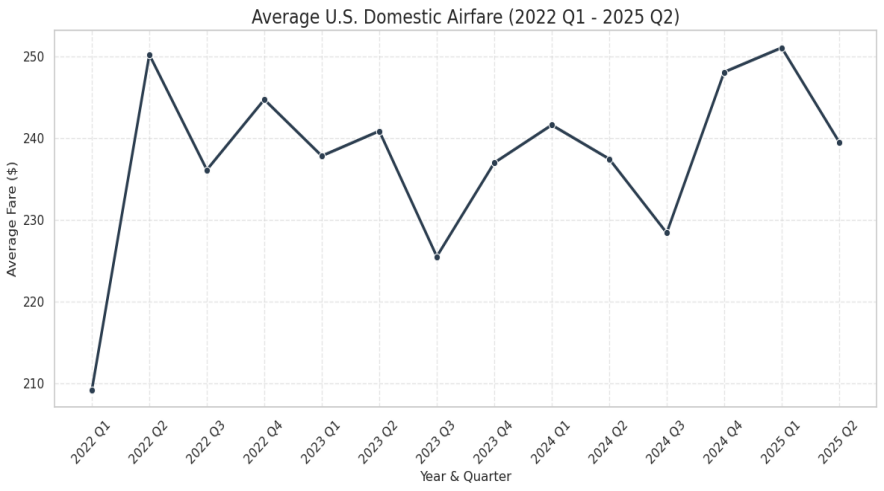
Our Gradient Boosting model achieved an **R^2 of 0.86**, with a Mean Absolute Error of only **$18.47**.



Key finding:

- Adding market structure variables increased R^2 from **0.4822** to **0.6815** — an improvement of **0.1993 (≈20 percentage points)**.
- Framed relative to the remaining explainable variation, market structure explains an additional **28 percentage points** of airfare variation.

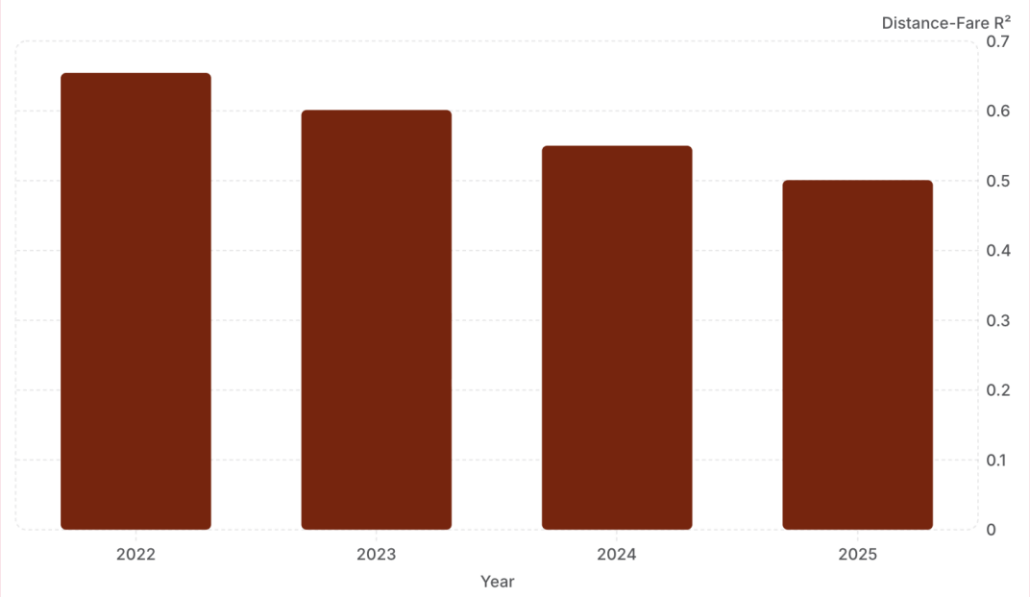
SEASONAL AIRFARE DYNAMICS



Key finding:

- Airfares exhibit a distinct cyclical pattern, typically peaking in **first half of the year**, most likely due to spring break and summer vacation surges.
- Prices then drop to their annual low in **Q3 (July–September)**, bottoming out at **$225** during the post-summer travel slump.
- While **Q1 (January–March)** represents the winter season with moderate fares, the unusually low prices of **$210** observed in early 2022 represent a significant statistical outlier, likely driven by COVID-19 reco**very** rather than standard seasonal demand.

THE DISTANCE-FARE CORRELATION IS WEAKENING



- The correlation between distance and average fare has dropped significantly over the past three years.
- This confirms that distance is becoming less predictive of airfare prices, while market power and competition dynamics are increasingly important.

STRATEGIC RECOMMENDATIONS FOR TRAVELERS

THE "STUDENT BUDGET HACK"

- **Distance ≠ Price:** Recognize that market structure dictates cost more than miles. A short monopoly flight (e.g., PHL to PIT) can cost 3x more than a long competitive route.
- **The "Hub-Skip" Strategy:** Avoid "Fortress Hubs" like Charlotte (+16% premium) or Atlanta (+12%). Flying from a nearby competitive airport can save a family of four over $400.

- **The 10% LCC Filter:** Only book legacy carriers (Delta, United) on routes where a Low-Cost Carrier has at least 10% market share; this "competition pressure" lowers legacy fares by 15–20%.
- **The Value Decision:** In competitive markets, the price gap between Legacy and LCC is often only ~$60. If checking bags, Legacy carriers usually offer better total value and reliability.

ACTIONABLE INSIGHTS FOR STAKEHOLDERS & POLICYHOLDERS

STAKEHOLDERS

- **Market Disruption Targets:** Use predictive simulations to identify routes overcharged by $100+ relative to distance and cost.
- **High-Potential Expansion:** Launching a Columbus (CMH) to Philadelphia (PHL) route presents a massive opportunity; fares are predicted to drop from ~$300 to $188, capturing immediate market share.
- **Data-Driven Pitching:** Airports in "Fortress Hub" shadows can use fare-premium data to prove the existence of a "Monopoly Tax," creating a guaranteed demand case for new entrants.

POLICYHOLDERS

- **Quantifying Welfare Loss:** High-dominance routes show a structural decoupling of price from operational costs, indicating clear market failure.
- **The "10% Threshold":** Data confirms that even minimal competition (10% share) acts as a binary switch for lower prices. Protecting the existence of small budget airlines is more effective than direct price regulation.
- **Merger Oversight:** Simulations predict that removing a Low-Cost Carrier from a market (e.g., via merger) causes an immediate price spike of ~$35 per ticket across the entire route.
- **Targeted Enforcement:** Regulators should prioritize oversight on the "Top 10 Rip-Off Routes" where consumer harm is most concentrated.

CITATIONS

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THANK YOU AND HAVE A SAFE FLIGHT!